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The returns to invention during the British industrial revolution†

By SEAN BOTTOMLEY*

It was a commonplace among contemporaries, and remains received wisdom today, that inventors were poorly remunerated during the industrial revolution. Adapting a dataset of 759 British inventors, this article presents the first large-scale attempt to examine the issue systematically. Using probate information, the article shows that inventors were extremely wealthy relative to the adult male population. Inventors were also significantly wealthier than another group who would have received a similar inheritance (in terms of both financial and social capital) and entered similar occupations: their brothers. Their additional wealth was derived from inventive activities: invention paid.

'It has done me harm, my friend. It has aged me, tired me, vexed me, disappointed me. It does no man any good to have his patience worn out, and to think himself ill-used.'

Daniel Doyce abandoning his invention, *Little Dorrit* (1855–7)[1]

In presenting Daniel Doyce as a poor, struggling, 'disappointed' inventor, Charles Dickens was invoking a well-worn literary trope that would have been familiar to his readership. At this time, inventors were almost invariably presented as 'the miserable victim of his own powerful genius'; 'the poor, honest, industrious, and hard-working mechanic', and 'the poor, honest, unprotected genius'; 'meritorious but unfortunate'; as 'Martyrs of Science' who worked 'alone, unfriended, solitary'; and who were 'mostly poor and of obscure condition, toiling unaided'; while 'the recorded instances of the[ir] martyrdom would be a task of enormous magnitude'.[2] Even Samuel Smiles, who exalted the achievements of inventors in his best-selling *Self-help* (1859), also observed how often their careers ended in disappointment, his hagiographies ending not 'with gold and glory, but with acrimony and theft'.[3] Intriguingly, the image of the poor, betrodden inventor was common throughout the industrializing west—in France, for example, Gustave Flaubert was moved to

*Author's Affiliation: Max Planck Institute for European Legal History.

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[1] Dickens, *Little Dorrit*, p. 509.

[2] In order: *London Journal of Arts and Sciences*, 1825, p. 316; *Mechanics' Magazine*, 1827, p. 324; French, *Life and times of Samuel Crompton*, p. 106; Timbs, *Stories of inventors*, p. viii; Hamilton Fyfe, *Triumphs of invention*, p. v; Woodcroft, *Brief biographies of inventors*, p. vii; Burnley, *Romance of invention*, p. 44. Most of these quotations appear in a longer discussion on the topic in Pettitt, *Patent inventions*, pp. 40–3, 126 (where she observes 'how pervasive and tenacious the stereotype of the poor artisan inventor proved to be').

[3] MacLeod, *Heroes of invention*, p. 256.

satirize the caricature in his *Dictionnaire des idées reçues*: 'INVENTORS: They all die in hospital. Another profits from their discovery, it is not fair'.[4]

However, this caricature cannot be disparaged as just another manifestation of Victorian maudlinism. There are multiple examples of inventors from the industrial revolution who, although responsible for important technical advances, ended their days in penury. Three examples are Henry Cort (1741?–1800), inventor of the puddling process for refining iron, who was ruined by the ill-timed death of a business partner; John Kay (1704–1780/1), inventor of the flying shuttle, who left for France after failing to enforce his patent in England; and Richard Trevithick (1771–1833), pioneer of high-pressure steam engineering, also frustrated by legal action. Instances such as these have meant that the Victorian view of the poor, financially disappointed inventor is still commonly accepted in the economic history literature. Mokyr, for example, notes that 'a large number of important inventors died in obscurity and poverty, indicating that the private returns to a socially useful invention were low'.[5] Similarly, in the social and cultural history literature, even those historians who have critically examined how Victorians (mis)represented and conceived inventors have also accepted the notion that they were rarely well recompensed. MacLeod, for example, notes at various junctures how the Great Exhibition threw 'into sharp relief the inadequacy of many inventors' rewards' and how 'it was an easy (and not inaccurate) strategy to portray the inventor as the victim of the state's ingratitude and the capitalist's ruthlessness'.[6]

The question of the private returns to invention, however, has not been systematically examined, with one partial exception: Harley's painstaking work on business accounts in the textile sector.[7] This indicates that those firms which adopted new technology first were quickly joined by other firms that served to reduce the first-movers' profits margins: supra-normal profits were rarely found in textiles after the early 1780s. Harley's conclusions, however, are inevitably constrained by the paucity of surviving accounts, recognizing that 'three firms are, of course, a very small sample and their profit figures may well not be representative of the industry' (albeit, with the suggestion that they probably overestimate the profit rate in the industry as a whole).[8]

The relative neglect of this issue is all the more surprising considering that whether inventors obtained financial rewards commensurate with their technical achievements has significant repercussions for how we might account for the origins of the industrial revolution—for which there are, broadly speaking, two competing explanations. The first can be characterized as a 'supply-side' explanation, attributing the industrial revolution to a pronounced increase in the supply of new ideas and concepts emerging in Britain and north-west Europe over the seventeenth and eighteenth-centuries. In this interpretation, these new ideas and ways of thinking were what initiated the development of industrial technology and revolution. McCloskey, for example, argues that perceptions of entrepreneurial activity began to improve during the eighteenth century and that as 'ordinary conversation about innovation and markets became more approving', both were

[4] Quoted in Baudry, 'Histoire de la propriété intellectuelle', p. 272.
[5] Mokyr, *Lever of riches*, pp. 158–9.
[6] MacLeod, *Heroes of invention*, pp. 212, 190.
[7] Harley, 'Technological change'.
[8] Ibid., p. 521.

able to flourish.[9] Mokyr also argues for the primary importance of the supply of ideas, stressing that although 'factor prices might have determined the *direction* of technological change, the *power* and *intensity* of improvements were a function of technological capabilities and motives that had deeper causes'.[10] These motives were not necessarily financial: 'when the decisions [to invent] were made largely by independent individuals, ambition, curiosity, and altruism may have had a larger role relative to naked greed'.[11] In light of the pervasive assumption that inventors were poorly remunerated, such an argument seems sensible.

The second can be seen as a 'demand-side', incentives-based explanation—that, simply put, it proved to be profitable to invent and commercialize the technology of the industrial revolution. This is not to imply that there is any unanimity on the reasons why the 'correct' incentives first appeared in Britain. Allen, for example, has emphasized the role of factor prices: in Britain, expensive wages combined with cheap energy and capital incentivized the development of technologies that substituted capital and fuel for labour which, he suggests, characterize the most important inventions of the industrial revolution.[12] These inventions could only have been developed and commercialized in Britain, as only there was it profitable to do so.[13] Conversely, new institutional economists such as North and Acemoğlu have emphasized the role of institutions in dictating incentives.[14] To take the example of the patent system as an institution, North argues that although the social rate of return to inventive activities has always been high, due to the non-rivalrous nature of invention, inventors without property rights in their inventive output will be unable to appropriate a private return therefrom.[15] Consequently, inventive activities will be spurned. However, with the institution of the patent system in England in 1624, inventors there were now accorded systemic property rights, increasing the private returns to inventions and contributing in turn to eventual industrialization.[16] Although subsequent work has shown that an effective patent system did not immediately emerge with the passing of the Statute of Monopolies in 1624, one did develop over the course of the eighteenth century.[17]

It is also worth pointing out that endogenous growth theories frequently assume that 'profit incentives determine the amount of research and development directed at different factors and sectors'.[18] Given the apparent lack of profit incentives for research and development, it would be doubtful whether endogenous

[9] McCloskey, *Bourgeois dignity*, p. 7.

[10] Mokyr, *Enlightened economy*, p. 272.

[11] Mokyr, 'Intellectual origins', p. 322. Elsewhere, however, Mokyr does acknowledge that 'Allen's basic assumption that inventive activity was driven by a desire to make money is not controversial'; idem, *Enlightened economy*, p. 270.

[12] Allen, *British industrial revolution*, p. 141.

[13] Allen provides a detailed illustration of this argument by examining James Hargreaves's spinning jenny. In brief, Allen argues that the spinning jenny could only have been invented in Britain because, in virtue of high labour costs, only there would it 'generate enough profit to make the cost of developing and perfecting [it] worthwhile'; Allen, 'Industrial revolution in miniature', p. 903. Without entering the minutiae of the debate, Allen's argument has been challenged on both theoretical grounds (Gragnolati, Moschella, and Pugliese, 'Spinning jenny'; for a reply, see Allen, 'Spinning jenny') and on empirical grounds (Styles, 'Fashion').

[14] North and Thomas, *Rise of the western world*; North, *Structure and change*; idem, *Institutions*; Acemoğlu and Robinson, *Why nations fail*.

[15] North, *Structure and change*, p. 164.

[16] North and Thomas, *Rise of the western world*, p. 154; Acemoğlu and Robinson, *Why nations fail*, p. 32.

[17] Bottomley, *British patent system*, pp. 170–4.

[18] Acemoğlu, 'Directed technical change', p. 781.

growth theory is applicable to the industrial revolution period; Crafts and Mills cautiously conclude after reviewing the applicability of growth models to the period that 'future attempts by growth economists to model the transition to modern economic growth should perhaps pay more explicit attention to improvements in capabilities and in incentive structures that increased the probability of technological advance'.[19]

Inevitably, the almost universal view that inventors were often disappointed financially has been invoked to query the evidential basis of these incentive-based arguments. Clark, for example, notes that 'the empirical difficulty with the appropriability argument is the appallingly weak evidence that there was any great gain in the returns to innovators in England in the 1760s and later . . . the Industrial Revolution economy was spectacularly bad at rewarding innovation'.[20] Clark uses the example of the cotton industry, where by the 1860s the value of innovations 'alone' was 'about £115 million in extra output per year [but] a trivially small share of this value of extra output ever flowed to the innovators'.[21] Clark concludes that 'the increased rate of innovation in Industrial England was the result not of unusual rewards but of a greater supply of innovation, still modestly rewarded'.[22] Two counter-points will suffice for the moment. One, it is still the case today that only a very small share of the economic and societal benefits of invention accrue to its progenitors. Nordhaus estimates that in the US, only 2.2 per cent of the total social returns to invention are appropriated by inventors; it is doubtful whether it would be possible (let alone desirable) for them to appropriate anything but a small share of the total benefits of their inventions.[23] Two, even if only a very small share of the untrammelled bounty of industrialization should accrue to an individual, it would still make them extremely wealthy (as we shall see).[24]

This article examines the returns to invention during the industrial revolution by using probate valuations of inventors' estates. In line with the assumption that technical success rarely promised financial success, we are specifically concerned with 'invention' rather than 'inventive activity'. Whereas the latter relates to the full gamut of activities involved with the production of new technology (for example, the conception of a new idea, its research and development, its commercialization),

[19] Crafts and Mills, 'From Malthus to Solow', p. 92.

[20] Clark, 'Great escape', p. 20.

[21] Ibid. A similar argument appears in Clark, *Farewell to alms*, pp. 230–58. There he also notes that of the 379 Britons who died in the 1860s with a probated wealth in excess of £500,000, only 17 worked in textiles; ibid., p. 236. This work is derived from Rubinstein, *Men of property*, pp. 60–7, who also observes that there were relatively few fortunes generated by manufacturing, compared with landed wealth and/or trade. This, however, overlooks what was truly remarkable: that great fortunes could be generated in manufacturing at all. In his classic survey of social mobility in the sixteenth and seventeenth centuries, Stone, 'Social mobility in England', p. 35, observes that for the mid-seventeenth century, the 'avenues of upward mobility are precisely those operating a hundred years later, in the middle of the eighteenth century: four fast elevators: marriage, the law, high government service, and the colonies; three medium fast: trade, government contracting and finance; and two slow: estate management, and professions other than the law'; manufacturing is omitted.

[22] Clark, *Farewell to alms*, p. 237.

[23] Nordhaus, 'Schumpeterian profits', p. 22.

[24] However, for the moment consider the three engineers Clark (*Farewell to alms*, p. 236) mentions during this discussion: Richard Trevithick, George Stephenson, and Humphry Davy, who 'though their names survive to history . . . captured very few of the social rewards their enterprise wrought'. All three appear in the list of inventors discussed in this paper—and while it is true that Trevithick died a poor man without probated wealth, Davy's wealth was probated below £30,000 (≈£27,500) on his death in 1829 (TNA, Prob 8/222) and Stephenson's below £140,000 in 1848 (≈£130,000; TNA, Prob 8/241). This would place them roughly in the top 0.5% and the top 0.05% of the adult male population respectively.

invention is defined here as, from a strictly technical perspective, the successful outcome of inventive activities (although we are obviously also concerned with inventors who sought to profit from their inventions, rather than 'pure' scientists). The article does not speak to the technically unsuccessful, be they the endeavouring but unlucky, or the deluded, peddling perpetual motion machines and the like.

The first section discusses the probate process in Britain during this period, establishing that it provides a tolerably accurate precis of an individual's wealth at death from 1796 onwards. The second section then discusses a list of over 700 inventors born between 1660 and 1830, first compiled by Meisenzahl and Mokyr, whose probates have been collated. A comparison with the overall distribution of probated wealth among adult males reveals that these inventors were in fact extremely wealthy. This wealth, however, may not have been derived from their inventions. It may have been accrued over the course of their 'normal' business career and/or inherited. The final section examines information from the family backgrounds of a subset of inventors who died between 1800 and 1870. In particular, it compares the probated wealth of inventors with those of their adult brothers, who tended to enter similar occupations and who would have enjoyed a similar inheritance (in terms of both social and financial capital). The article establishes that inventors were significantly wealthier than their brothers, even if they were 'talented'. Invention paid.

I

In England and Wales before 1858, the established church was responsible for the probate of wills, the process whereby the legality of the will was confirmed and approval granted to the executors to carry out the instructions of the deceased, or, in those cases where the deceased had not prepared a will, the granting of an administration for the estate. This arrangement stemmed from the medieval period when literacy was largely confined to the clergy and over time it developed into a highly complex system with multiple overlapping jurisdictions. Whenever someone died leaving behind movable goods worth more than £5 (termed *bona notabilia*, literally 'noteworthy goods'), their will would usually be probated in the local diocese or peculiar court. When the deceased left behind *bona notabilia* in multiple dioceses, then the estate would either be probated in the Prerogative Court of York (PCY, covering northern England), or in the Prerogative Court of Canterbury (PCC, covering the midlands, southern England, and Wales). If there was *bona notabilia* in both archdioceses, then the estate was usually probated at the PCC (although occasionally assets in each archdiocese were probated in their respective prerogative court). Also, even if the entirety of their estate was in one diocese, the wealthiest tended to use the PCC, making it the pre-eminent probate court in England and Wales.[25] This byzantine arrangement risked making the task of finding probated estates (and, crucially, the probate valuations) extremely time-consuming although, as we shall see, the majority of inventors were wealthy

[25] English, 'Probate valuations', p. 81. The records of the various commissary and sheriff courts are now held at the National Archives of Scotland and are searchable online at https://www.scotlandspeople.gov.uk/advanced-search#{"category":"legal","record":"legal-wills-testaments"}, (accessed on 8 Dec. 2016). Fifty-three inventors had wills proved in Scotland.

enough so that they had *bona notabilia* in multiple dioceses, meaning that their estates could usually be found in the PCC, and/or the PCY. Another complication was the maintenance of an entirely separate system of probate courts in Scotland. Before 1824, secular commissary courts had been responsible for the proving of wills, but thereafter, responsibility was transferred to the local sheriff courts (of which there were 32 in 1830).

As part of the probate process, executors were responsible for compiling an inventory of the testator's estate and, from 1796 onwards, a figure for its total value. In the case of the PCC, these are available in the Probate Act Books, as is the case with the PCY and other diocese courts.[26] Probate figures can also be found in the death duty registers,[27] which is especially useful for the minor diocese and peculiar courts where the probate may not be so readily found. Because the coverage of death duty registers is national (and so includes all English and Welsh peculiar, diocesan, and archdiocesan courts), it might have been preferable to rely on the death duty registers for all 1796–1857 probates. Unfortunately, they are not always complete. For example, of 1,749 wills proved in the PCC in 1800, only 41 per cent also appear in the death duty registers.[28] Probate valuations become much simpler to find from 11 January 1858, when the Court of Probate Act came into force.[29] The Act transferred the responsibility for probate to a civil court, while also establishing the Principal Probate Registry which produced an index of all estates so probated.

In one vital respect, probate valuations can be relied upon. The executors who prepared the probate had to swear an affidavit that they had carried out their task faithfully and, especially after 1796 when Pitt the younger introduced a full-scale tax on inheritances, the Inland Revenue checked the figures with which they were provided. Fraudulent behaviour was punished, fines levied, and, although contemporaries complained about almost every other aspect of the byzantine system outlined earlier in this section, evasion was thought to be rare.[30] Conscientiously prepared they may have been, but probates did not provide a precise net valuation of the entire estate belonging to the deceased. The key difficulties, in ascending order of importance, are: an exact monetary figure was not provided until 1881, they omitted debts owed by the deceased, and they did not include real estate.

Proceeding in this order, before 1881 probate valuations were not recorded as precise figures but were instead entered as being under some upper-bound limit—for example, below £10,000, below £9,000, below £8,000, and so on. These probate valuations were statutory, and so we can be confident that an estate valued below £10,000 was worth approximately £9,500. By 1815, there were 50 separate probate categories, with the lowest being below the value of £20 (on which no

[26] The Probate Act Books for the PCC are available at TNA, Prob 8. The Probate indexes for the PCY are available at the Borthwick Institute, York, classmark YDA/11. Most of the PCY indexes are also available online, as are those for the larger diocese courts such as Chester and Durham, on genealogical websites such as findmypast.com and ancestry.co.uk.

[27] TNA, IR 26.

[28] Green and Owens, 'Metropolitan estates', p. 299.

[29] 20 & 21 Vict. c. 77.

[30] Green, Owens, Maltby, and Rutterford, 'Lives in the balance?', p. 317.

estate duty was to be paid) and the highest category being reserved for those with a probated value in excess of £1,000,000 (with an estate duty of £15,000).[31]

Concerning debts, previous work indicates that they rarely consumed a significant proportion of the estate, especially for those who were nearing the end of their working lives or had already retired (as was usually the case with the inventors examined for this study—for the 621 inventors with certain dates, the average life span was 69 years; only 14 per cent died before the age of 55).[32] In a sample of 1,439 estates probated from 1870 to 1902, debts, on average, accounted for only 10.8 per cent of the gross estate.[33] In a similar study for the period 1810–40, debts appear to be more significant. A comparison between 39 wills that were probated in the PCC to a value in excess of £2,000, with the full accounts of the deceased deposited at the Legacy Duty Office, shows that in seven instances, debts consumed over half of the estate.[34] Nonetheless, this still means that probates do provide an accurate indication of the net value of the large majority of estates in this period.

Another potentially serious matter is the exclusion of land from the probate values. The scale of the problem can be assessed for those inventors who were alive in 1873 by tracing their land holdings in the parliamentary *Return of Owners of Land*.[35] The *Return* is especially useful as it provides not only the acreage of land belonging to each owner, but also the gross annual rental income from which the actual value of the holding can be estimated. In line with contemporary opinion, a multiplier of 15 years gross rental was applied to urban land and a multiplier of 30 years gross rental to rural land.[36] Every inventor who was resident outside London (excluded by the *Return*), and who died in the 15 years between 1874 and 1888, was searched for in the *Return* (70 in total; those who died after 1888 were excluded because there was an ever-increasing chance that their land holdings in 1873 would not be an accurate indication of their actual land holdings upon decease). In 54 instances (77 per cent), land holdings were equivalent to less than a quarter of their eventual probate; 40 held no land at all. This, however, still leaves 16 instances where land was worth more than a quarter of their probate and eight instances where land holdings were either approximately the same in value as their

[31] Stamp Act 1815, 55 Geo. III, c. 184. The bands did vary slightly over time. For more details, see English, 'Probate valuations', p. 82.

[32] Compared with overall life expectancy at birth (35.9 years in 1800), this average lifespan may appear to be very high, although the more appropriate comparison is with life expectancy in early to mid-adulthood (say 30, an age which most inventors had to attain before they had acquired the technical expertise to begin inventive activities), which between 1750 and 1799 was 62.1 years for males; Wrigley and Schofield, *Population history of England*, p. 250.

[33] Green and Owens, 'Geographies of wealth', p. 857.

[34] However, the more modest the probated value of the estate, the more likely it was that debts would account for a greater proportion of it. Of 59 estates probated below £2,000, debts consumed over half of the estate in 17 cases. The figure for estates in excess of £2,000 is provided in the main text though, because, as the following section shows, the majority of inventors' wills were probated above this figure; Owens, Green, Bailey, and Kay, 'Measure of worth', p. 402.

[35] Great Britain, *Return of Owners of Land 1873* (P.P. 1874, C.1097); idem, *Return of Owners of Land and Heritages* (P.P. 1874, C.899).

[36] Although the Parliamentary Return does not state whether the land was urban or rural, it is usually straightforward to gauge which one it was from the gross annual rental income per acre, high values (£100+) denoting urban land, lower values rural land. This is line with the rates of return used in Green et al., 'Geographies of wealth', p. 854.

probate, or indeed greater than it.[37] John Foster, for example, a worsted spinner responsible for introducing exotic fibres such as alpaca and mohair into textile production, left behind an estate probated at below £250,000 (≈£225,000). He also held a substantial estate of 3,684 acres centred on Hornby Castle in north Lancashire, producing a gross annual rental income of £5,792 (which, assuming this was primarily agricultural land, implies a value of approximately £173,760). This means that there are a minority of cases where using the probate figures alone will seriously understate the gross value of an individual's assets.

On an individual basis, probate valuations are imperfect—especially as they exclude land holdings. From the analysis of land holdings in 1873, we can see that probate will significantly understate an inventor's wealth at death in around one-quarter of cases. The exclusion of debts from probate is unlikely to be as important (due in part to the relative longevity of inventors) and Owens et al. conclude 'that in the large majority of cases, [probate] valuations are a reasonably good estimate of true net worth, a finding that helps to underpin other research which has used probates to assess wealth distributions'.[38]

II

Meisenzahl and Mokyr have recently constructed a database of 759 British inventors born between 1660 and 1830.[39] Their primary concern was to include not only the 'great inventors' of the period but also 'tweakers, engineers and mechanics who made minor improvements on existing inventions'.[40] To this end, the database was constructed using not only standard prosopographical sources such as *The Oxford dictionary of national biography* and A. W. Skempton's *Biographical dictionary of civil engineers* but also a wide range of industrial case studies.[41] Also, to be included in their database, an individual had to be responsible not only for technical advances worthy of note in the literature, but also with its commercialization: 'they had to have made some inventions themselves ... but their main activity was implementation'.[42] This means that capitalists and manufacturers who may have funded or adopted new technology are not included, unless they were also inventors in their own right. Similarly, pure scientists are also excluded.

This database is eminently suitable for the purposes of this article: it has a wide chronological and industrial coverage, and it reaches beyond the more prominent inventors to include those whose (accumulated) efforts certainly contributed as much to British industrialization. Some changes have been made to the original list of inventors used by Meisenzahl and Mokyr. In particular, those inventors

[37] This result suggests a slightly higher propensity for inventors to hold land than other manufacturers. Rubinstein, '"Gentlemanly capitalism"', p. 163, finds that less than 15% of wealthy businessmen who died in 1873–5 (defined as those who left behind £100,000+) held more than 20% of their wealth in land. Thompson, *Gentrification*, pp. 69–71, also finds that only 12% of businessmen with fortunes less than £50,000, owned any form of real estate.

[38] Owens et al., 'Measure of worth' p. 401.

[39] Meisenzahl and Mokyr, 'Rate and direction of inventive activity'.

[40] Ibid., p. 454.

[41] On the methodological dangers of relying exclusively on traditional prosopographical sources, see MacLeod and Nuvolari, 'Pitfalls of prosopography'.

[42] Meisenzahl and Mokyr, 'Rate and direction of inventive activity', p. 454.

who emigrated before the end of their careers have been omitted (this accounts for only 18 inventors in the original series). This is largely because this article is specifically concerned with investigating the appropriability of returns to invention in the British economy, but also because foreign wills will not have been subject to the same valuation method as British probate. For similar reasons, Britain is defined here as the island of Great Britain, that is, excluding Ireland, which did not undergo industrialization during this period (with the exception of parts of Ulster), but including Scotland (which did industrialize).[43] Finally, where in the light of additional biographical information an inventor did not meet the original Meisenzahl and Mokyr criteria, they have been excluded. 'Horace Hall', for example, was actually a pseudonym used by two Frenchmen, Cachard and Lanthois, when taking out a patent in 1814 for a spinning frame for flax.[44] This brings the total used here down to 707 from an original 759.

The remaining 707 inventors should not be regarded as a definitive list of all inventors who meet the criteria used here and who made non-trivial advances. One inventor who could have been included is John Lethbridge (1675–1759) who invented the first underwater diving machine in 1715.[45] A better-known example might be Isambard Kingdom Brunel (1806–59), who amongst his many engineering triumphs pioneered the use of wrought iron for bridge-building, use of the screw-propeller in shipping, and the tunnelling shield (although his father was also responsible for the latter). However, no additions have been made to the list—otherwise it might appear that inventors have been 'cherry-picked' for the exercise undertaken here. It might also have been possible to expand the list to include more female inventors (only one, the remarkable Eleanor Coade, appears in Meisenzhal and Mokyr).[46]

Of these 707 inventors, 547 (77.4 per cent) either left behind a will disposing of their estate or, more unusually, failed to prepare a legal will and so passed on an estate that needed to be administered. Moreover, for those inventors who died during the eighteenth century, the absence of a will or an administration does not automatically entail poverty—they may have already disposed of their assets prior to death or felt certain that informal arrangements would suffice, or quite simply the records are missing. For example, the Devon ironmonger Thomas Newcomen amassed a small fortune in royalties for his atmospheric engine, estimated at between £5,000 and £10,000 on his death in 1729.[47] Unfortunately, there is no indication that his estate was probated in the PCC, and most Devon wills for this period were destroyed by bombing in 1942.

More detail becomes available from 1796 onwards, when probate figures are usually noted on the will itself or in a probate act book maintained by the Court (or failing that, probate figures can also be found in the Death Duty Registers produced by the Inland Revenue). Table 1 divides the wealth of 422 inventors (from our 707) who died between 1800 and 1870 into five categories: those who left behind less than £200 or no probate at all, those who left behind less than £1,000 (that is,

[43] Also, most Irish wills were destroyed in by the Four Courts Fire in 1922 meaning the probate valuations are unobtainable.
[44] Rimmer, *Marshalls of Leeds*, p. 171.
[45] Davis, *Deep-diving*, p. 583.
[46] Kelly, 'Coade'.
[47] Bottomley, *British patent system*, pp. 246–7.

Table 1. *Probated wealth of inventors, 1800–70*

Probated wealth	*Adult male population (1839–41)*	*Adult male population (1858)*	*Inventors*[a]
<£200 or no will	73,302 (88.14%)	87,043 (87.70%)	124 (29.4%)
£200–£999	5,570 (6.70%)	6,690 (6.74%)	39 (9.2%)
All <£1,000	94.84%	94.44%	163 (38.6%)
£1,000–£9,999		4,554 (4.59%)	104 (24.6%)
£10,000–£49,999	4,296 (5.16%)	812 (0.82%)	95 (22.5%)
£50,000+		154 (0.16%)	60 (14.2%)
All >£1,000	5.16%	5.56%	259 (61.4%)

Notes: *a* Includes inventors probated in Scotland, whereas the figures for adult male population relate only to England and Wales. Considering that Scotland industrialised soon after England, though, it is unlikely that its distribution of probated wealth was significantly different.
Please refer to app. I for details of how the distribution of male probated wealth was estimated for 1839–41 and 1858.
Sources: See section I.

between £200 and £999 19s. 11d.), those leaving behind between £10,000 and £50,000 (described as 'very prosperous' by Rubinstein) and those leaving more than £50,000 ('definitely rich').[48] This sub-period has been chosen for four reasons (besides the availability of probate figures). First, the careers of these inventors would have coincided roughly with the classical period of the industrial revolution dated between *c.* 1760 and 1830. Second, this was a period of modest inflation (except for the final years of the Napoleonic Wars). Third, Rubinstein reports that the distribution of probated wealth remains relatively constant until around 1870, when more adult males start to be probated in the higher categories.[49] Finally, it is possible to confirm Rubinstein's observations about wealth, as for four years in this period it is possible to collate evidence for the overall distribution of adult male probated wealth: 1839–41 and 1858.[50]

It is immediately apparent that relative to the adult male population, inventors were extremely wealthy. Whereas roughly 5 to 6 per cent of the adult male population had their wealth probated in excess of £1,000, more than 60 per cent of inventors left behind more than £1,000. The contrast becomes greater as we move upwards through the wealth distribution: less than 1 per cent of adult males left behind more than £10,000, whereas almost 40 per cent of inventors did so. Unfortunately, the probate data do not exist to extend this exercise far back into the eighteenth century, although there is one strong indication that inventors in this earlier period were also relatively wealthy. Of the 95 English inventors who died between 1700 and 1799, 57 had estates that were probated in the two archdiocese courts (exactly three-fifths), which, for reasons discussed earlier, were where the wealthiest tended to enter their wills. By contrast, this was at a time when only about one-sixth of men were probated in any archdiocese, diocese, or peculiar court.[51]

Clearly not every inventor reaped the rewards of their technical achievements—around 30 per cent either had probated wealth of less than £200 or no probated

[48] Rubinstein, *Men of property*, p. 151.
[49] Ibid.
[50] For details on these estimates, please see app. I.
[51] Clark, 'Consumer revolution', p. 8.

Table 2. *Probated wealth of inventors as a multiple of average nominal earnings, 1796–1905*

	0–4	*4–40*	*40–400*	*400–4,000*	*4,000+*
1796–1840	66	31	53	41	9
1840–66	49	28	60	52	11
1867–1905	29	24	51	73	23

Source: The data for average nominal earnings are from L. H. Officer and S. H. Williamson, https://www.measuringworth.com/ukearncpi/ (accessed on 8 Dec. 2016).

wealth at all. We are also excluding those unfortunates who felt driven to try their luck abroad (although, as already mentioned, this only accounts for 18 inventors of the original 759). However, the proportion of inventors leaving either very little probated wealth, or none at all, does diminish over time, and the proportion of wealthy inventors increases: the industrial revolution did witness an improvement in the appropriability of invention.

Table 2 divides the 600 inventors who died between 1796 and 1905 into three cohorts of 200 each, according to their date of death (1796–1840 for the first 200 and 1867–1905 for the final 200). To facilitate the comparison over time, their probate values were recalculated as a multiple of the average annual nominal earnings in the year of their death and then placed into five categories: those with probated wealth equivalent to between zero and four times the average annual nominal earnings, those with between four and 40 times the average earnings, up to the extremely wealthy leaving behind more than 4,000 times the average annual earnings.[52] These categories are not arbitrary. Before 1809, the top probate category was 'upper value', which was defined as those leaving behind more than £100,000. Unfortunately, no other detail was given relating to the exact value of the probate, so because £100,000 was roughly equal to 4,000 times the nominal wage in the 1800s, it was chosen as the top wealth category for this exercise.

In each successive cohort, the number of 'poor' inventors leaving behind less than four times the average annual wage (≈£85 in 1796, ≈£275 in 1905) declines, as does those leaving behind less than 40 times average annual earnings. Conversely, the overall number of inventors in the top two categories increases: the returns to invention were increasing throughout the industrial revolution, over and above increases in average earnings.

III

It does not, however, automatically follow that the wealth of inventors was actually derived from their inventions. These were presumably talented individuals and their income may have been accrued over the course of a 'normal' business career and/or inherited. Similarly, the results in table 2 could be interpreted as a function of the rising costs of invention: as technology became increasingly sophisticated, so

[52] Data on average annual earnings is from L. H. Officer and S. H. Williamson, https://www.measuring worth.com/ukearncpi/ (accessed on 8 Dec. 2016).

the increasing costs of research and development meant that only those who were already relatively wealthy could engage in inventive activities.

If we consider first the possibility that inventors may have been the beneficiaries of financial inheritances, this is not a straightforward issue to explore. For every inventor of the stature of Richard Arkwright in this study, there are several who are as equally obscure. One such would be Squire Diggle. We know from his entry in Day and McNeil's *Biographical dictionary of the history of technology* (itself based on an 1878 industry study of weaving), that Diggle was responsible for several improvements to power looms, including a drop box facilitating the weaving of different colours; that he obtained several patents between 1845 and 1858; and that he worked in Bury.[53] Building on this information, it has been possible to flesh out a little more of his life using probate and genealogical services (helped in this case by his unusual name). It is likely that he was the Squire Diggle born in Birtle in the parish of Bury in 1803 to John and Mary Diggle and that he died nearby in Whitefield (also in the parish of Bury) in 1868, with probate valued at below £50,000 (≈£47,500). As difficult as it has been to trace these sparse details of Squire Diggle's life, the problems are only multiplied when considering his family. His father, for example, is almost entirely inscrutable. We know from Squire Diggle's baptism that his father was called John, but it does not record his occupation. Neither is there any record of a John Diggle leaving behind a probated estate. All that can be reasonably inferred is that Squire Diggle was not born of privilege.

It might be thought that the best way to investigate the issue of inheritance would be to track down the wills of fathers and see how much they left behind to their (inventive) son. However, as discussed previously, wills and probates become more difficult to find during the eighteenth century. Also, an inheritance may not have come from the direct male line. Instead, where possible, as a proxy for the social background of an inventor, the occupation(s) of their father has been tracked down—and this was possible for 499 different inventors. The occupations of these fathers have then been coded using the same Primary, Secondary, Tertiary (PST) occupational classification scheme used by the Cambridge Group for the History of Population and Social Structure to prepare their census of adult male employment for England and Wales in 1817.[54] The PST classification scheme is highly detailed; table 3 compares the overall adult male employment for 1817 with those of inventors' fathers for the primary, secondary, and tertiary sectors and for selected occupations within each of these sectors.

It is apparent that the family background of these inventors was atypical. In 1817, 39.4 per cent of the adult male population worked in the primary sector and 36 per cent in agriculture. The equivalent figures for the family backgrounds of our 499 inventors are 12.4 per cent and 10 per cent.[55] This difference becomes starker when we bear in mind that the inventors in this exercise were born between 1660 and

[53] Day and McNeil, *Biographical dictionary*, p. 361; Barlow, *History and principles of weaving*, p. 52.

[54] The full classification scheme is available to download at Cambridge Group for the History of Population and Social Structure, 'Occupational coding: the PST system', http://www.campop.geog.cam.ac.uk/research/projects/occupations/britain19c/pst.html (accessed on 3 April 2018).

[55] There are (frequent) occasions when a father had more than one occupation. It is also common to find a father's occupation denoted inconsistently—for example, in one source he might be described as a 'shipowner' and in another as a 'merchant'. In either instance, both occupation codes were allotted, but with a half value.

Table 3. *Occupational family background of inventors born 1660–1830 (%)*

Occupational sector	*c. 1710*	*1817*	*Fathers of inventors*
Primary	50.8	39.4	12.4
Secondary	37.2	42.1	52.0
Tertiary	12.0	18.4	35.6
Detailed breakdown			
Primary			
Agriculture		36.0	10.0
Mining and quarrying		3.1	1.9
Other primary		0.3	0.5
Secondary			
Food industries		2.8	1.2
Footwear		3.7	0.8
Textiles		7.7	8.1
Wood industries		2.0	3.9
Instrument making		0.3	3.6
Chemical, soap, adhesives		0.1	0.9
Iron and steel manufacture		3.0	8.5
Machines and tools		1.1	14.3
Building and construction		7.1	2.2
Other secondary		14.3	8.5
Tertiary			
Dealers, unspecified		0.6	3.6
Dealers in textile products		0.2	2.2
Domestic service		2.2	0.0
Professions		1.3	11.0
Armed forces		1.2	2.3
Distinguished, titled, gentleman		1.3	1.8
Other tertiary		11.6	14.7

Note: The detailed breakdown of fathers' occupations includes just over 75% of fathers.
Sources: Figures for occupational sector are from Shaw-Taylor and Wrigley, 'Occupational structure', p. 59. Figures for the detailed breakdown are from Kitson, Shaw-Taylor, Wrigley, Davies, Newton, and Satchell, 'Creation of a "census"', pp. 38–40.

1830—and clearly the comparison year of 1817 is at the very end of a period which saw significant changes in the occupational distribution of adult males. The next earliest comparable figures for primary, secondary, and tertiary employment are from *c.* 1710 and these have also been included in this table (unfortunately, detailed employment shares for *c.* 1710 are not yet available). If we were to take 1710 as our comparison year, then the disparity between the overall adult male population engaged in agriculture (be they labourer, yeoman, or gentleman farmer) and the family backgrounds of our inventors would be yet more pronounced.

As would be expected, most inventors had family origins in the secondary sector (52 per cent). The secondary sector, however, incorporated a vast range of trades. It is noteworthy that comparatively few inventors had family backgrounds in occupations that experienced only limited technical change (represented in table 3 by the food industries, footwear, and building and construction). By contrast, those with family origins in what has been previously termed the 'labour aristocracy' are over-represented. In particular, more inventors' fathers were in the machines and tools sector (14.3 per cent) than in the entire primary sector. Put another way, the son of an adult male working in machines and tools was over 40 times more likely to produce an advance that would merit inclusion in this dataset of inventors than the son of a worker in the primary sector. Finally, there is perhaps a surprisingly high proportion of inventors with backgrounds in the tertiary sector (35.6 per cent).

This is partly because the division between tertiary and secondary sectors might not always be so clear-cut—some of the dealers in textile products, for example, probably had a hand in manufacturing at some point in their career and would have been able to provide their son with an entrée into the sector. However, it is also because professional, university-educated men tended to have well-educated children themselves (20 inventors had fathers who were clergymen, for example), and they appear to have gravitated towards higher technology sectors such as chemicals. One example of this is William Hyde Wollaston (1766–1828). The son of a Norfolk vicar, he graduated from Cambridge and discovered a new method of manufacturing malleable platinum (a method he kept secret, to his considerable profit). Unfortunately, just over a quarter of fathers' occupations could not be traced. It can be safely asserted that they were not of the titled classes or gentry—whose genealogies are readily available in publications such as *Burke's peerage* or *Burke's gentry*—but with this caveat, there is no reason to believe that there would be a marked difference between those inventors' fathers who have proved traceable and those that have not, to alter radically the results in table 3.

The table has little to say directly on the importance of the inheritance of financial capital—although scions of the wealthiest landed classes appear in nearly the same proportion as they would have done in the overall population; the inheritance of financial capital from their father does not appear to be as significant as the inheritance of social capital, at least in terms of becoming a noteworthy inventor. At the other end of the social scale, not a single inventor is recorded as having grown up in a foundling hospital or a workhouse or a similar institution. This is consistent with other work by Clark examining the correlation of wealth between 147 father-and-son pairs from the first half of the seventeenth-century. Rich fathers (as measured by their probated wealth) tended to have rich sons. However, if the economic advantage of the sons was primarily derived from the size of the bequest they had received from their father, then sons with more siblings would have seen their potential bequest dissipated among their brothers and sisters. However, the number of siblings had little influence on how wealthy sons were at death. Rich fathers tended to have rich sons, no matter how many there were. Clark concludes that 'the principal advantages the fathers were transmitting to their sons were thus either cultural or even genetic'.[56]

It is also possible to use other evidence from the families of inventors to show more convincingly that invention did indeed pay. The final part of this article compares the probated wealth of inventors with those of their adult brothers. This is a particularly apposite group for comparison. First of all, as we have just seen, inheritance mattered—perhaps not always in terms of financial inheritance, but certainly in terms of the social milieu in which children grew up. Non-inventive brothers presumably enjoyed the same advantages in this respect as their inventive brothers. In a second related point, brothers often entered similar occupations. It is these similarities that explain why 24 of the inventors in the database were related as brothers—the talents and opportunities required to become an inventor were clearly not evenly distributed among the adult male population. We can thus factor out inheritance and occupation in accounting for systemic differences in probated wealth between inventors and their brothers.

[56] Clark, *Farewell to alms*, p. 120.

Table 4. *Wealth of brothers of inventors, 1800–70*

	Probated wealth of inventor					
	<£200 (25)	*<£1,000 (11)*	*<£10,000 (35)*	*<£50,000 (44)*	*£50,000+ (28)*	
Probated wealth of brother						Totals
<£200	31	12	26	35	23	127 (50.2%)
£200–£999	3	3	7	7	2	22 (8.7%)
£1,000–£9999	9	2	14	31	13	69 (27.3%)
£10,000–£49,999	2	2	3	9	8	24 (9.5%)
£50,000+	–	–	3	2	6	11 (4.3%)

Note: Tab. 4 uses the same wealth categories as in tab. 1 to compare the wealth of inventors (distributed in the top row of the table), with those of their brothers (distributed in the main body of the table). For instance, of the 25 inventors who left behind wealth probated at less than £200, it has been possible to confirm 45 brothers who reached adulthood and died in Britain between 1800 and 1870 as well. 31 of these brothers also had wealth probated at less than £200.
Sources: See section I.

The obscurity of many of the inventors used in this article has already been emphasized and reliably tracing out the life course of their brothers, most of whom are entirely unremarkable, is even more difficult. Two strategies have been used. First, the *Oxford dictionary of national biography* and the other publications used by Meisenzahl and Mokyr in the compilation of their list of inventors occasionally include their family details (usually parentage, but sometimes siblings as well). Second, details for other brothers have been found thanks to the availability of birth, marriage, and death certificates, as well as census records, on genealogical websites such as ancestry.com and findmypast.com. Family trees are also sometimes available on familysearch.org (compiled by the Church of Latter Day Saints, in accordance with its doctrine of 'Baptism for the dead') and on ancestry.com (compiled by family genealogists)—although these are often unreliable and have been used primarily as a finding aid.

For 143 of the 422 inventors discussed in table 1, it was possible to confirm the existence of at least one adult brother who reached at least the age of 25 and who died in Britain between 1800 and 1870 (253 brothers in total). Cases where both brothers were inventors have been omitted as well as those cases where the non-inventive brother(s) were in partnership with the inventive brother or there is a record of a significant bequest among the siblings (although the latter rarely occurred; nieces and nephews tended to be the beneficiaries when a bequest was made to another branch of the family). Even so, it is nigh-on certain that there were many more brothers who could have been included in this exercise, but unfortunately the information needed to establish whether a brother had a probated estate (for example, their date of death and/or their residency) is simply untraceable.

Despite these limitations in data collection, it is still possible to show that inventors tended to be much wealthier than their brothers. Of the 143 sets of brothers, the inventor was the wealthiest in 102 instances (71 per cent). In only 25 instances (17 per cent) was there at least one other non-inventive brother who was richer. In the remaining cases, there was at least one other brother of equal wealth (or, more frequently, the inventor and his brother(s) did not have any probated wealth at all). A more detailed illustration of the wealth of inventors in comparison to their brothers is provided in table 4. The top row divides the 143 inventors with brothers into the same wealth categories as those used in table 1, with the number in parentheses denoting how many of the 143 inventors are in each category. The distribution of wealth for these 143 is roughly the same as in table 1, although here there are relatively fewer inventors in the lowest wealth category (<£200) and relatively more in the highest wealth category. The columns beneath this then show the distribution of the wealth of their brothers. So, there are 25 inventors in this exercise whose estate was either probated at less than £200, or with no estate to probate at all. Of their 45 brothers, 31 also had either no probated wealth or less than £200. Three had probated wealth between £200 and £1,000, nine between £1,000 and £10,000 and two between £10,000 and £50,000 (one of these two had the good fortune to inherit a dukedom). None left behind more than £50,000. At the other end of the table, the 28 inventors who left behind more than £50,000 had 52 brothers whose wealth could be identified. Only six of their brothers were also able to achieve this category of wealth—23 had little or no probated wealth to speak of at all.

Overall, if inventors were wealthier than their brothers, then the latter should be concentrated at the top and to the right of the table, and away from the bottom left corner. Clearly, they are—overwhelmingly so when one considers how important simple happenstance can be in influencing an individual's financial success over the course of their career. To take one example of an inventor used in this exercise, Peter Ewart (1767–1842) was an engineer who developed early steam turbines. Although he came from a fairly obscure background (his father was a Church of Scotland minister in a small Dumfriesshire village), he did moderately well, leaving behind an estate probated at below £8,000 (≈£7,500).[57] His brother William (1763–1828), however, had the good fortune to enter into partnership with Sir John Gladstone at an early stage in his career. Sir John became the largest of the Liverpool slave traders.[58] When William died in 1828, his probate was valued at below £180,000 (≈£170,000). Similarly, fortunes could be made and squandered in the same lifetime. William Fothergill Cooke (1806–1879), for example, was an early pioneer of the electric telegraph and co-founder of the Electric Telegraph Company. Selling his patent to the newly founded company netted him £90,000 as well as a substantial share-holding in the new company. Everything, however, was lost in slate mining in Wales, and on his death in 1876 he left behind £16.[59]

However, by inventing successfully, the inventors *ipso facto* constitute a talented population who would presumably have been well remunerated even if they had followed alternative career paths. The same cannot be assumed for their brothers, and so a subset of brothers who are likely to have been talented have been chosen for a final comparison. The subset is composed of the 153 brothers who can be positively identified as entering a profession (for example, military officer, lawyer, doctor, surgeon, engineer, or merchant), being engaged in highly skilled manual work (for example, clockmaker, engineer, or optician), large-scale manufacturers

[57] Cardwell, 'Ewart, Peter'.
[58] William Ewart also became godfather to one of Sir John's children: William Ewart Gladstone, the prime minister. Farrell, 'Ewart, William'.
[59] Burnley, 'Cooke, Sir William'.

Table 5. *Wealth of talented brothers of inventors, 1800–70*

	Probated wealth of inventor					
	<£200 (15)	*<£1,000 (7)*	*<£10,000 (25)*	*<£50,000 (32)*	*£50,000+ (18)*	
Probated wealth of talented brother						Totals
<£200	12	4	17	17	10	60 (39.2%)
£200–£999	2	3	5	4	2	16 (10.5%)
£1,000–£9999	5	1	12	19	11	48 (31.4%)
£10,000–£49,999	1	2	2	7	7	19 (12.4%)
£50,000+	–	–	2	2	6	10 (6.5%)

Sources: See section I.

who owned or co-owned their own business,[60] and/or those who enrolled at Oxbridge.

Table 5 is organized in the same way as table 4, and a preliminary inspection re-affirms the conclusion that invention paid: even with the omission of 'untalented brothers', the remaining talented brothers are concentrated in the top right-hand corner of the table. Similarly, in 66 of the 97 exclusively talented brother sets, the inventor was the wealthiest at death (67 per cent). Moreover, this exercise will understate the difference between inventors and their talented brothers. Some of the latter will have been excluded from this exercise because there is no evidence to identify them as talented (at least, according to the criteria adopted here). This problem would tend to affect the poorer ones who are less well documented.

Finally, birth order does not appear to have been an unduly significant influence on whether a (male) child would become an inventor. Of our 143 inventors, 59 had one other brother who reached adulthood (and this includes those brothers who died outside the 1800 to 1870 period used for table 4). In 34 instances, the inventor was the first born; in 25 instances, the second born. Similarly, in the 36 instances where there were three brothers, the inventor was the first born in 14 instances, the second born in 10 instances, and the third born in 12 instances.

IV

Previous work has relied primarily on impressionistic evidence to assert that inventors in this period rarely obtained financial rewards commensurate with their technical achievements; the introduction noted some prominent examples. This article, however, has sought to examine the issue systematically, using a large database of inventors and the probate valuation of their estate at death. First of all, it has shown that when compared with the adult male population, inventors were extremely wealthy. It is worth reiterating that many of the inventors studied in this article are (rightly) obscure, and were responsible for incremental changes ('microinventions'). The poor artisan inventor was increasingly atypical and Flaubert's cynicism concerning the trope was justified. It has also shown

[60] Specifically, manufacturers who had five or more employees (information that appears in the 1851 census) and/or manufacturers who subsequently adopted an honorific such as 'esquire' or 'gentleman'.

how, relative to average nominal wages, the returns to invention were gradually improving over the course of the industrial revolution. If it were possible to extend the exercise presented in table 2 further back into the eighteenth and seventeenth centuries, it would probably show that this trend had been going on for some time. While inventors were frequently probated in the archdiocese courts during the eighteenth century (a strong indication of a significant wealth holding), there are no prominent examples of an inventor dying in the seventeenth century having made his fortune. The reasons for this improvement have not been discussed, although speculatively it may be attributable to broad improvements in the institutional setting, such as the patent system, and/or increases in the scale of production, in turn increasing the number of marketable units to which a prospective advance could be applied.[61]

Unfortunately, direct measurement of the income generated from inventions (assuming that this could be neatly distinguished from their 'normal' business income anyway) is in most cases precluded by the lack of surviving accounts, as illustrated by the work of Harley. However, an approximation can be achieved by comparing the probated wealth of inventors with their brothers, who were in all essential respects the same (they would have enjoyed a similar inheritance and they tended to enter similar occupations), except that they were not themselves inventors (or at least, sufficiently inventive to be included in Meisenzahl and Mokyr's list). That inventors were significantly wealthier than their brothers (talented or otherwise), especially when we bear in mind the countervailing effects of happenstance and luck on the results in tables 4 and 5, indicates that it was invention itself which paid. In conjunction with previous work that has demonstrated how inventors responded to market signals during this period, that invention was remunerated would offer strong empirical support for the demand-side, 'incentives' explanation of the industrial revolution.[62]

A final shibboleth was discarded when discussing the backgrounds of inventors. Their families were not especially poor (or rich), but inventors did emerge from highly atypical backgrounds, particularly the 'labour aristocracy' and the professions: the image of the prosperous, upwardly mobile inventor is slightly less romantic than the spurned working-class genius, but at least it has evidence to recommend it. This also indicates the importance of social milieu, background, and access to ideas—itself supporting a 'supply-side' account of the industrial revolution. It is likely that a more persuasive account of the origins of the industrial revolution will be one that does not make a hard distinction between the 'demand' and 'supply' of new ideas and technology, but one that offers a clearer explanation of how both sides of this market interacted with one another.

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[61] In a process akin to the one detailed by Sokoloff, 'Inventive activity'.
[62] Bottomley, 'Patenting in England'.

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Appendix I: Compiling the adult male distribution of probated wealth

For four years in the period under consideration (1839–41 and 1858), it is possible to obtain information on the overall number of probates and estate administrations according to their values from Parliamentary Papers.[63] Beginning with 1858 (the year the Court of Probate Act came into force), the estates of 20,867 males were either probated or administered (another 9,112 estates belonged to females, but as all bar one of the inventors in this exercise were male, their estates are a more apposite group for comparison than the overall adult population). The next stage was to estimate the number of adult male deaths in England and Wales for the year (that is, those aged 21 years or more at death; those aged 20 or less could not prepare legal wills). The *Twenty-First Annual Report* notes that there were 227,220 male deaths recorded in 1858.[64] The male deaths are not divided by an exact age of death, but in five- or 10-year windows (for example, 10–14 or 15–24). By adding up the number of deaths in those windows where the deceased was an adult, as well as two-fifths of the number of deaths recorded for the ages between 15 and 24 (as an estimate for the number of deaths from ages 21 to 24), we arrive at a figure of 102,049

[63] Great Britain, *Probates and Administrations* (P.P. 1843, XXX); idem, *Twenty-Second Annual Report* (P.P. 1861, XVIII), pp. 174–81.
[64] Great Britain, *Twenty-First Annual Report* (P.P. 1860, XXIX), pp. 98–9.

deceased adult males for 1858. Then, because the probate court only came into operation on 11 January, this figure was multiplied by 355/365 (to omit the extra 10 days; 1858 was not a leap year), to produce a final estimate of 99,253 as the number of adult male deaths in England and Wales from 11 January to 31 December 1858, who could have been probated in the new Court. We can then use the probate data to estimate, for example, that the 154 adult males probated in excess of £50,000 would have represented 0.16 per cent of the recorded adult male deaths for 1858. Repeating this with the next three lower wealth categories leaves us with a remainder of 87,043—that is, those adult males with estates valued at less than £200.

This same methodology has been repeated for 1839–41, but with some minor adjustments. First, probates and administrations for these years are not, unfortunately, categorized between male and females. However, because males accounted for 74 per cent of estates probated in excess of £1,000 in 1858 (as well as 74 per cent of those probated between £200 and £1,000), it has been assumed that the same proportion of such estates also belonged to males in 1839–41.[65] So, for example, of the 17,433 estates probated in excess of £1,000 in these three years, approximately 12,889 would have belonged to males, or around 4,296 per annum.[66] Second, the number of male deaths in 1839, 1840, and 1841 is available in the Registrar General's *Ninth Annual Report* (1848).[67] Unfortunately, information on age at death is only available for 1841 as a calendar year (previously this information was provided by years ending on 30 June), but repeating the same exercise outlined above for 1858 to calculate adult male deaths in 1841, and assuming the same ratio of adult to infant deaths pertains to 1839 and 1840, produces an estimate for the number of adult male deaths for each year.[68] Once this estimate was prepared, it was possible to calculate the probated wealth distribution of adult males for these three years, which table 1 presents on an annualized basis.

[65] For probates valued at less than £200, this figure falls to 67%.

[66] Unfortunately, the Parliamentary Paper which provides the probates for 1839–41 does not subdivide wealth categories above £1,000.

[67] *Ninth Annual Report* (P.P. 1848, XXV), p. 41. Unlike the case with 1858, there is a minor discrepancy between the number of deaths recorded by the Registrar-General and Wrigley. In 1841, for example, total deaths registered were 322,680, compared with Wrigley's revised figure of 329,134. Although Wrigley can be regarded as definitive, the Registrar-General's figures are preferred here. This is for three reasons: the Registrar-General figures are available for Wales as well as England (matching the same area covered by the probate data), most of Wrigley's additional deaths would have been of infants, and the Registrar-General figures are available by age. Wrigley and Schofield, *Population history of England*, p. 636.

[68] So for 1841, of 174,198 deaths: 71,595 were below the age of 5; 9,093 aged 5–9; 4,478 aged 10–14; 5,604 aged 15–19; and 1,327 aged 20 (one-fifth of the 6,633 deaths at ages 20–24), leaving 82,101 deaths at 21 or above; *Fifth Annual Report* (P.P. 1843, XXI), p. 32. Assuming the same ratio of adult to infant deaths, this means that there were 81,426 adult male deaths in 1839 (from a total number of 172,766 male deaths) and 85,977 adult male deaths in 1840 (from a total of 182,421).